

AN INTERACTIVE FIELD GUIDE

Precision has a floor.

The Cramér–Rao Bound connects what measurements reveal to what estimators can achieve.

From repeated noisy measurements to Fisher information, bias, and the geometry of localization.

[Start with the experiment →](#)

SAME TRUTH · LESS SPREAD

 $n = 4$

SD 1.00

 $n = 16$

SD 0.50

 $n = 64$

SD 0.25

True μ

$$\text{Var}(\bar{X}) = \sigma^2 / n$$

An estimate is one outcome. Precision is a distribution.

Keep the unknown parameter fixed. Repeat the entire measurement process.

01

A fixed world

The unknown value θ does not change between trials.



02

New measurements

Each trial produces a fresh random dataset X .



03

A new estimate

Apply the same rule $T(X)$. Its spread is the variance.

THE ESTIMATOR

$$T = T(X), \quad \mathbb{E}_{\theta}[T] = \theta$$

Unbiased means correct *on average*, across repeated datasets.

THE QUESTION

How narrow can that cloud be?

Not “How close was this single estimate?”

Not “How small is the observation noise?”

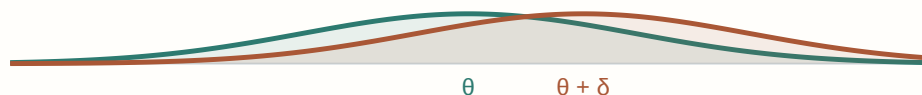
Nearby worlds are hard to tell apart.

Information measures how sensitively the data distribution changes when the parameter moves.

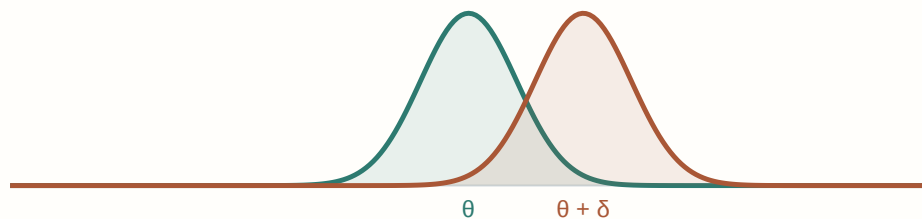
SAME SEPARATION · DIFFERENT NOISE

Could these observations tell you which world is true?

Noisier observations



Cleaner observations



These are **data densities**, not posterior probabilities.

FROM AMBIGUITY TO PRECISION

Small changes must leave a detectable trace.

If θ and $\theta + \delta$ generate almost indistinguishable data, a locally unbiased estimator cannot be arbitrarily concentrated.

$$I_n(\theta) = \mathbb{E}_\theta \left[(\partial_\theta \log p(X; \theta))^2 \right]$$

Fisher information turns local sensitivity into a number. Larger information permits a lower variance bound.

Likelihood → score → Fisher information.

The likelihood is a function of the candidate parameter, with the observed dataset held fixed.

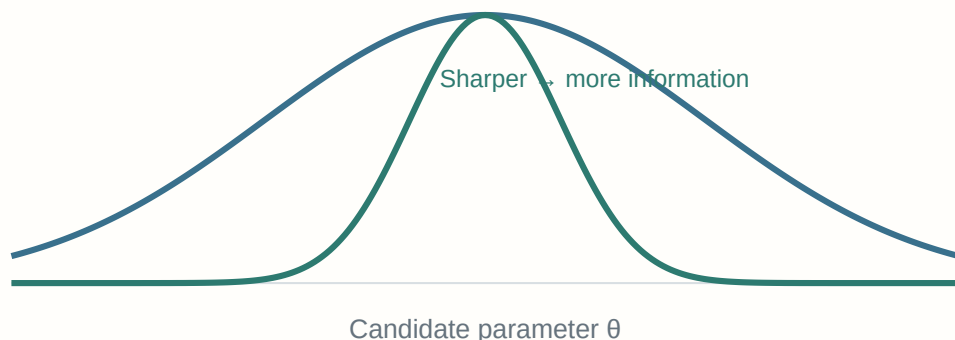
1 · LIKELIHOOD

How compatible is a candidate?

$$\ell(\theta; X) = \log p(X; \theta)$$

A likelihood is not automatically a probability distribution over θ .

Likelihood / maximum



2 · SCORE

Which way does the evidence push?

$$U_{\theta}(X) = \partial_{\theta} \ell(\theta; X)$$

Under regularity, $\mathbb{E}_{\theta}[U_{\theta}] = 0$.

3 · FISHER INFORMATION

Expected squared score = expected curvature

$$I_n(\theta) = \mathbb{E}_{\theta}[U_{\theta}^2] = -\mathbb{E}_{\theta}[\ell''(\theta; X)]$$

The curvature identity requires additional differentiability and interchange conditions.

The scalar Cramér–Rao Bound.

A floor for the variance of locally unbiased estimators in a regular model.

THE BOUND

$$\text{Var}_\theta(T) \geq \frac{1}{I_n(\theta)}$$

$$\text{SD}_\theta(T) \geq \frac{1}{\sqrt{I_n(\theta)}}$$

Variance has squared parameter units.
Standard deviation has parameter units.

READ THE FINE PRINT

The theorem has a scope.

Unbiasedness: $E\theta[T] = \theta$ in a neighborhood.

Regularity: differentiate under the expectation; the score has mean zero.

Finite information: $0 < I_n(\theta) < \infty$; T has finite variance.

Known model: the likelihood describes the actual experiment.

One score identity. One Cauchy–Schwarz inequality.

An unbiased estimator must move, on average, at the same rate as the true parameter.

THE SCORE IDENTITY

Differentiate unbiasedness:

$$1 \quad \partial_{\theta} \mathbb{E}_{\theta}[T] = 1$$

Move the derivative inside:

$$2 \quad \mathbb{E}_{\theta}[TU_{\theta}] = 1$$

Use $\mathbb{E}_{\theta}[U_{\theta}] = 0$:

$$3 \quad \text{Cov}_{\theta}(T, U_{\theta}) = 1$$

CAUCHY–SCHWARZ

Correlation cannot exceed one.

$$\text{Cov}(T, U_{\theta})^2 \leq \text{Var}(T) \text{Var}(U_{\theta})$$

$$1 \leq \text{Var}(T) I_n(\theta)$$

$$\boxed{\text{Var}(T) \geq 1/I_n(\theta)}$$

The entire lower bound follows from how strongly an unbiased estimator must respond to the score.

For a Gaussian mean, averaging is already optimal.

Assume independent measurements $X_i \sim N(\mu, \sigma^2)$, with σ known.

COMPUTE THE INFORMATION

$$\ell(\mu) = C - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$$

$$U_\mu = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)$$

$$I_n(\mu) = \frac{n}{\sigma^2}$$

COMPARE WITH AN ESTIMATOR

The sample mean meets the bound exactly.

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad \mathbb{E}[\bar{X}] = \mu$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \text{CRB}$$

No asymptotic approximation is needed.

For this model, equality holds for every $n \geq 1$.

What will happen when you collect more measurements?

In the next slide, every histogram bar comes from genuinely simulated datasets.

TRY THESE THREE MOVES

Predict → change → explain

16 → 64 samples: what happens to variance and standard deviation?

Double σ : which quantity gets four times larger?

Use only X_1 : does collecting unused data improve that estimator?

REFERENCE EXPERIMENT

Independent Gaussian measurements

16	2	1
samples / trial	noise σ	true μ

$$I_n = 4, \quad \text{CRB} = 0.25, \quad \text{SD}_{\min} = 0.5$$

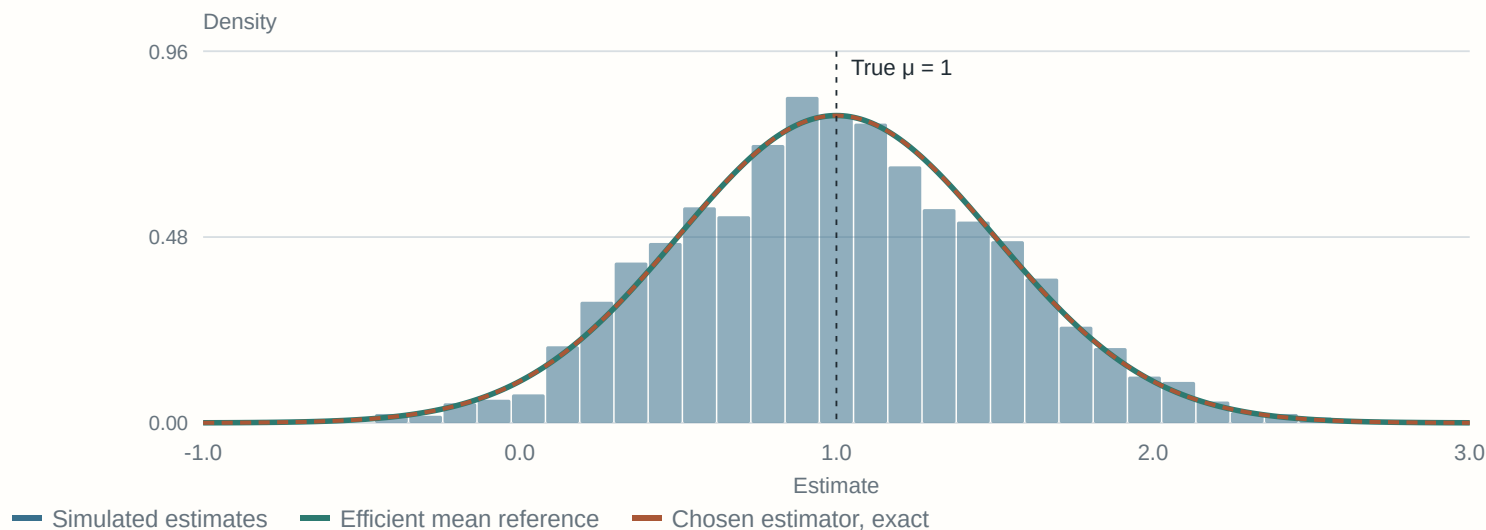
The truth is fixed across trials. Randomness comes only from new observations.

[Open the experiment →](#)

Can you reach the precision floor?

Change the experiment, repeat it, and compare empirical spread with the exact bound.

SAMPLING DISTRIBUTION · ONE ESTIMATE PER TRIAL



EMPIRICAL VARIANCE

0.2570

across repeated trials

EXACT ESTIMATOR VARIANCE

0.2500

theoretical, not fitted

UNBIASED CRB

0.2500

σ^2 / n , using all data

2,000 trials · seed 42 · empirical bias 0.0016 · 0 estimates outside plot. Each trial draws 16 new measurements.

A simulated variance slightly below the CRB is Monte Carlo fluctuation, not a violation.

INDEPENDENT $N(\mu, \sigma^2)$ DATA

Samples per trial, n 16

Measurement noise, σ 2.0

Estimator

Sample mean · use all data

Random seed

42

+1 trial

+1,000 trials

Reset

Exact efficiency: 100%. SD floor: 0.500. The sample mean attains the bound.

Truth $\mu = 1$ stays fixed.

Parameter changes replay 2,000 trials from the chosen seed.

Four times the data. Half the standard deviation.

Information adds for independent measurements; variance and standard deviation scale differently.

GAUSSIAN SCALING · $\Sigma = 2$



THE USEFUL ARITHMETIC

$$I_n = nI_1, \quad \text{CRB} \propto n^{-1}, \quad \text{SD}_{\min} \propto n^{-1/2}$$

N	VARIANCE BOUND	SD BOUND
4	1.0000	1.00
16	0.2500	0.50
64	0.0625	0.25

The iid shortcut fails for correlated or non-identical data. Use the information of the **joint** likelihood.

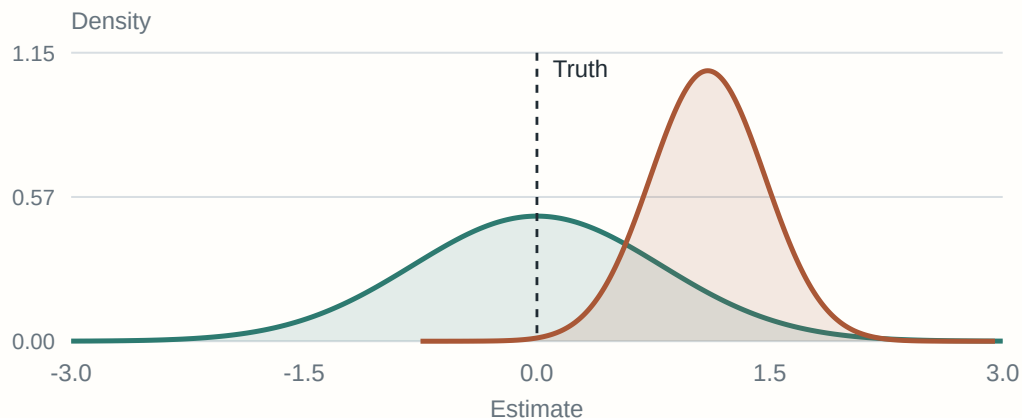
A narrow cloud can be centered in the wrong place.

The unbiased CRB is not a universal lower bound on mean squared error.

BIAS-VARIANCE DECOMPOSITION

$$b(\theta) = \mathbb{E}_{\theta}[T] - \theta$$

$$\text{MSE}_{\theta}(T) = \text{Var}_{\theta}(T) + b(\theta)^2$$



THE CORRECT BIASED EXTENSION

The derivative of the bias matters.

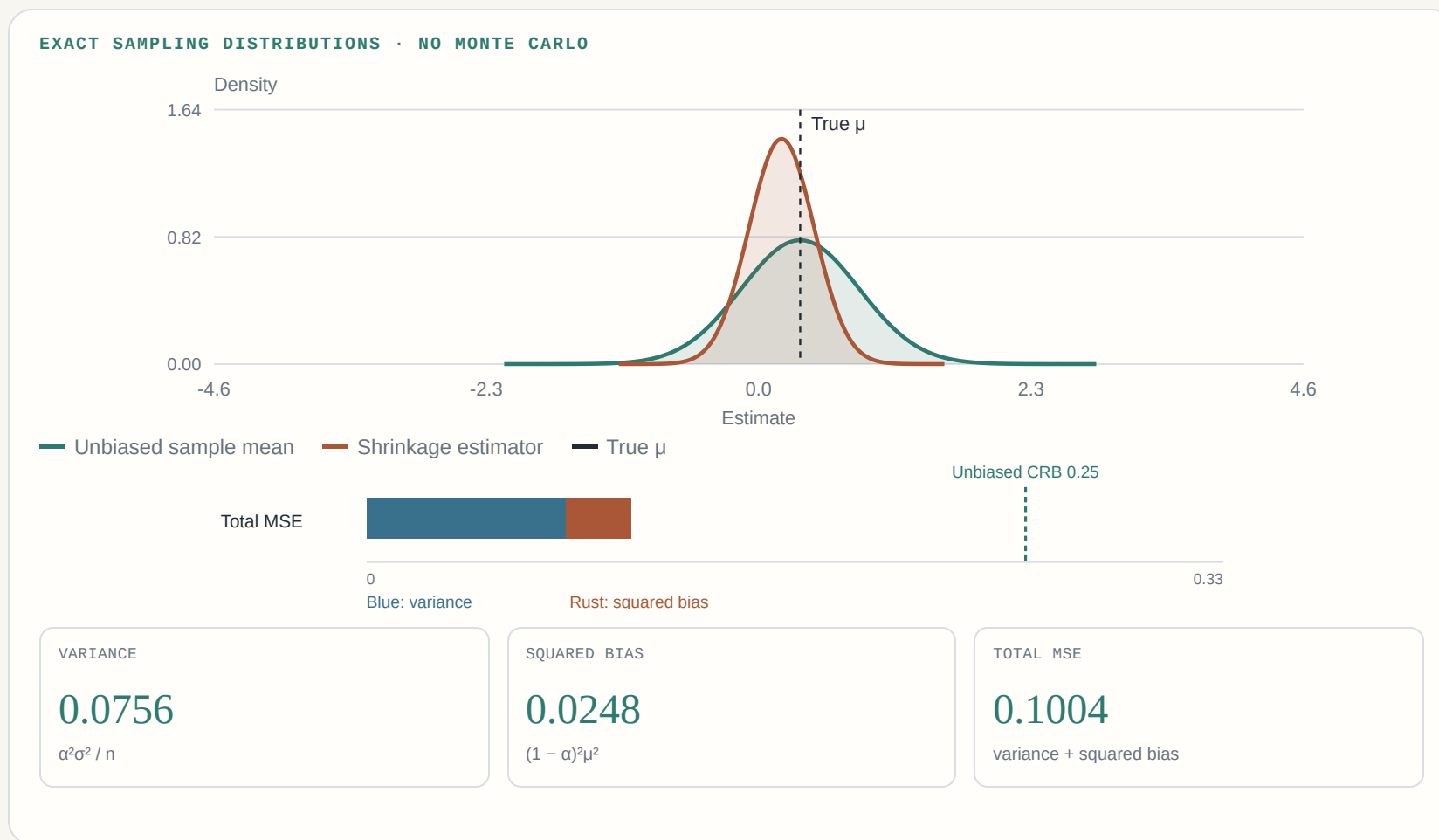
$$\text{Var}_{\theta}(T) \geq \frac{[1 + b'(\theta)]^2}{I_n(\theta)}$$

$$\text{MSE}_{\theta}(T) \geq \frac{[1 + b'(\theta)]^2}{I_n(\theta)} + b(\theta)^2$$

This follows from the same proof, replacing 1 by $\partial_{\theta}\mathbb{E}_{\theta}[T]$.

Can you “beat” the bound by shrinking toward zero?

Watch the variance fall, then move the true mean away from the shrinkage target.



TA = A $\bar{X}$ · TARGET = 0

Keep-data factor, α 0.55

True mean, μ 0.35

Near target

Far from target

No shrinkage

Always zero

Reset

$n = 16$ and $\sigma = 2$, so the unbiased variance bound is **0.25**.

MSE is below 0.25 here. Bias changes which bound applies.

Biased variance bound = $\alpha^2 / I_n = 0.0756$.

This estimator attains that bound, too.

Fixed α ; no prior is assumed. A lower MSE here is a pointwise bias–variance tradeoff, not a violation of the unbiased CRB.

“Lower bound” does not mean “always achievable.”

Equality requires a very specific relationship between the estimator and the score.

EQUALITY IN CAUCHY-SCHWARZ

All estimation error aligns with the score.

$$T(X) - \theta = \frac{U_\theta(X)}{I_n(\theta)} \quad \text{almost surely}$$

For uniform attainment, this must define the **same statistic $T(\mathbf{X})$** throughout the parameter range.

Gaussian mean: $\frac{U_\mu}{I_n} = \overline{X} - \mu$

THREE DISTINCT CLAIMS

Do not collapse them into one.

Unbiased: expectation equals the target.

Efficient: variance attains the CRB.

MLE: maximizes the likelihood; it may be biased at finite n .

Regular MLEs often approach the bound asymptotically. Finite-sample equality needs its own check.

Information has directions.

A vector parameter has a covariance matrix—not a single uncertainty number.

MATRIX CRB

The vector score is $\mathbf{u} = \nabla_{\theta} \log p(\mathbf{X}; \theta)$.

$$\mathbf{J}(\theta) = \mathbb{E}[\mathbf{u}\mathbf{u}^T]$$

$$\text{Cov}(\hat{\theta}) \succeq \mathbf{J}(\theta)^{-1}$$

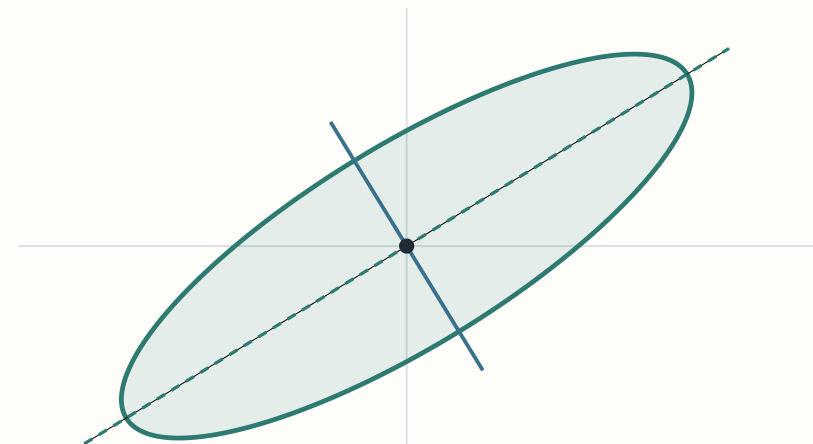
$$\mathbf{q}^T \text{Cov}(\hat{\theta}) \mathbf{q} \geq \mathbf{q}^T \mathbf{J}^{-1} \mathbf{q}$$

The inequality holds in every direction $\mathbf{q}$.

Assume $\mathbf{J}$ is nonsingular and the vector estimator is locally unbiased.

EIGENVECTORS TELL THE STORY

Covariance-bound ellipse



Weak information → long error axis

Strong information → short error axis

Large information eigenvalue → small variance bound along that eigenvector.

One range measurement constrains one local direction.

A simplified 2D ranging model: known anchor positions, independent Gaussian range noise.

THE MEASUREMENT MODEL

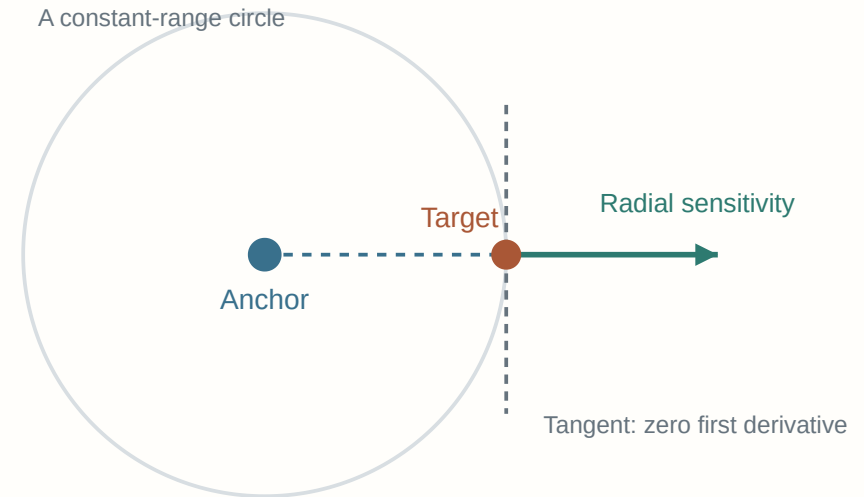
$$z_i = \|\mathbf{p} - \mathbf{a}_i\| + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_i^2)$$

$$\mathbf{u}_i = \frac{\mathbf{p} - \mathbf{a}_i}{\|\mathbf{p} - \mathbf{a}_i\|}$$

$$\mathbf{J}(\mathbf{p}) = \sum_i \frac{1}{\sigma_i^2} \mathbf{u}_i \mathbf{u}_i^\top$$

Each rank-one term adds information along a line of sight. Here $\mathbf{u}_i$ is a **unit direction**, not the score vector.

LOCAL SENSITIVITY



A small radial move changes the range to first order. A tangential move does not.

The target must not coincide with an anchor. Noise variance is assumed known and independent of position.

Same sensors. Different geometry. Different limits.

Four accurate ranges can still leave one position direction poorly constrained.

TRY THESE CONFIGURATIONS

What should the uncertainty ellipse do?

Surround: place anchors in different directions.

Cluster: move them to almost the same line of sight.

Collinear: place anchors and target on one straight line.

Then turn on an unknown common range offset. How much position information is lost?

THE SCALAR SUMMARY

Position error bound

$$\text{PEB} = \sqrt{\text{tr}(\mathbf{J}^{-1})}$$

For an unbiased position estimator:
root mean squared Euclidean error $\geq$ PEB.

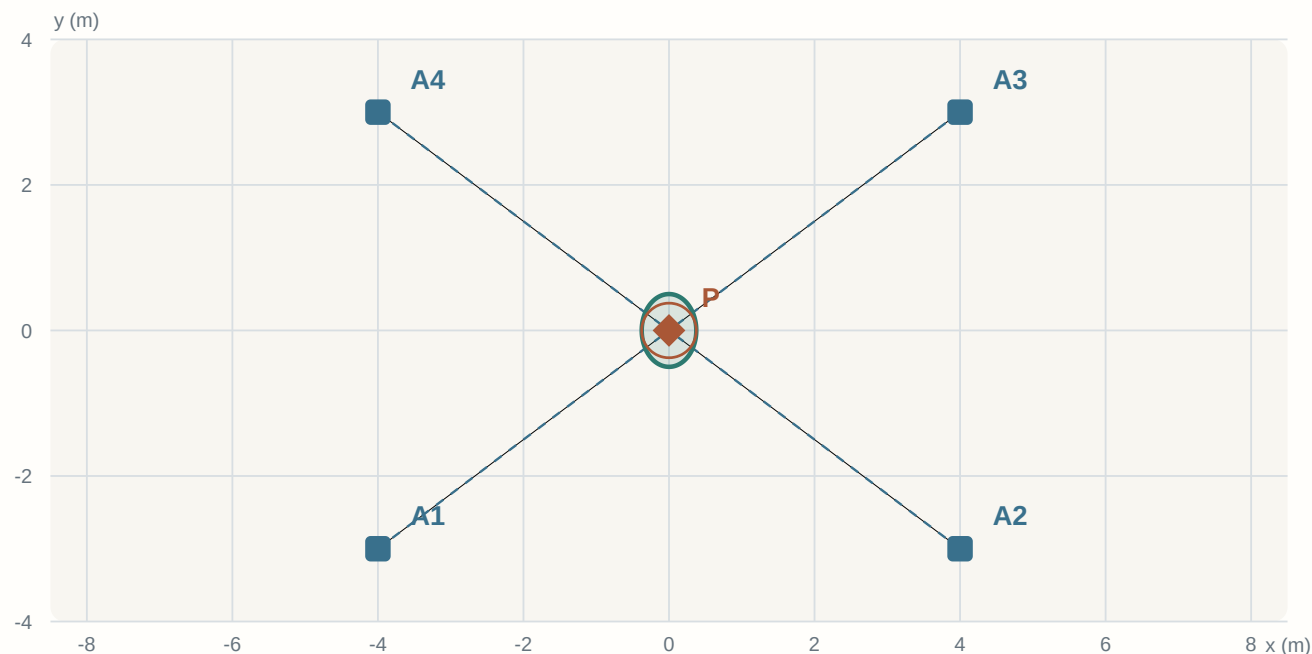
A finite ellipse describes **local** information. It does not rule out distant mirror solutions.

Move the sensors →

Move the sensors. Reshape the bound.

Drag any anchor or the target. The Fisher matrix and its inverse update immediately.

KNOWN ANCHORS · LOCAL INFORMATION AT THE TRUE TARGET



— Anchors A1–A4 — True target P — CRB ellipse

Unit-Mahalanobis CRB ellipse; not a confidence region. Geometry is evaluated at the true target.

Surround

Cluster

Collinear

Range noise, σ (m) **0.60**

☐ Also estimate a common range offset β

Select / move without dragging

Target P ▼

x (m) **0.00**

y (m) **0.00**

$J \text{ [m}^{-2}] = \begin{bmatrix} 7.11 & 0.00 \\ 0.00 & 4.00 \end{bmatrix}$

PEB = 0.625 m · condition 1.8

Information eigenvalues: 7.111, 4.000 m^{-2} . Common offset is known.

Reset all

An unknown clock offset consumes position information.

The same measurements must now explain both the target position and a common range offset β .

PARTITION THE INFORMATION

$$\theta = \begin{bmatrix} \mathbf{p} \\ \beta \end{bmatrix}, \quad \mathbf{J} = \begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{b}^\top & c \end{bmatrix}$$

$$\mathbf{J}_e = \mathbf{A} - \mathbf{b} c^{-1} \mathbf{b}^\top$$

The position block of $\mathbf{J}^{-1}$ is $\mathbf{J}_e^{-1}$, **not** $\mathbf{A}^{-1}$. Assume the required blocks are invertible.

INTERPRET THE SUBTRACTION

Some parameter effects can imitate each other.

If moving the target changes all ranges similarly, an unknown offset can explain part of that same change.

The Schur complement removes information that cannot distinguish position from the nuisance parameter.

$$\mathbf{J}_e \preceq \mathbf{A}$$

No loss when the coupling $\mathbf{b}$ is zero. A large loss is possible for clustered lines of sight.

[Toggle the offset in Lab 3 →](#)

A moving support breaks the familiar proof.

Not every model allows differentiation to pass through the expectation.

A USEFUL WARNING EXAMPLE

$X_i \sim \text{Uniform}(0, \theta)$

$$p(X; \theta) = \theta^{-n} \prod_{i=1}^n \mathbf{1}\{0 < X_i < \theta\}$$

$$U_\theta = -\frac{n}{\theta} \quad \text{inside the support}$$

$$\mathbb{E}_\theta[U_\theta] = -\frac{n}{\theta} \neq 0$$

The boundary moves with θ . The omitted boundary contribution is exactly why the score argument fails.

DO NOT FORCE THE SHORTCUT

Check the estimator directly.

$$T = \frac{n+1}{n} X_{(n)}, \quad \mathbb{E}_\theta[T] = \theta$$

$$\text{Var}_\theta(T) = \frac{\theta^2}{n(n+2)}$$

This variance is obtained from the sample maximum's distribution—not from the invalid regular CRB calculation.

Other warning signs: boundaries, nondifferentiability, singular information, and misspecified noise.

Design the experiment, not only the estimator.

A new independent, correctly modeled measurement adds a positive-semidefinite information term.

LINEAR GAUSSIAN MEASUREMENTS

$$\mathbf{z} = \mathbf{H}\boldsymbol{\theta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$$

$$\mathbf{J} = \mathbf{H}^T \mathbf{R}^{-1} \mathbf{H}$$

H controls sensitivity and geometry.

R controls noise and correlation.

Both matter. Counting measurements alone is not enough.

CHOOSE A DESIGN OBJECTIVE

Different questions favor different designs.

DESIGN GOAL

ONE COMMON CRITERION

Small total variance

Minimize $\text{tr}(\mathbf{J}^{-1})$

Small ellipsoid volume

Maximize $\log \det \mathbf{J}$

Protect the weakest direction

Maximize $\lambda_{\min}(\mathbf{J})$

Use a well-defined parameterization and units. These are local design criteria at the assumed true state.

Read a CRB result like a careful reviewer.

Before comparing an algorithm with a bound, make sure they solve the same statistical problem.

01 · ESTIMATOR CLASS

Is the comparison valid?

Measure or analyze bias. A learned, constrained, or regularized estimator is not automatically unbiased.

02 · PARAMETER VECTOR

What was treated as known?

Include unknown offsets, calibration, orientation, or other nuisance parameters before inverting.

03 · METRIC AND MODEL

Are the units and assumptions aligned?

Match variance with variance, MSE with MSE, and RMSE with a square-root bound. Model correlations correctly.

04 · LOCAL VERSUS GLOBAL

What can the bound miss?

Inspect rank, conditioning, competing solutions, and low-information regimes. A small local CRB is not a guarantee of successful localization.

Three ideas to keep.

More information lowers the bound. Bias changes the comparison. Geometry decides the weak directions.

01 · SENSITIVITY

Information belongs to the experiment.

$$I_n = \mathbb{E}[U_\theta^2]$$

The same estimator can face very different limits under different noise or measurement designs.

02 · PRECISION

Unbiased variance has a floor.

$$\text{Var}(T) \geq I_n^{-1}$$

A better algorithm cannot beat this floor while remaining in the theorem's estimator class.

03 · DIRECTION

The matrix tells you where.

$$\text{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{J}^{-1}$$

Independent, complementary measurements are often more valuable than repeated views of the same direction.

Continue from the bound to the bigger picture.

The equations and examples are teaching material; the sources below give the wider statistical context.

PRIMARY REFERENCES

S1 [Stanford STATS 200 · Lecture 15](#)

Fisher information, Cramér–Rao proof, efficiency, and nuisance-parameter intuition. Autumn 2016.

S2 [Ly, Marsman, Verhagen, Grasman & Wagenmakers \(2017\)](#)

A Tutorial on Fisher Information. arXiv:1705.01064.

S3 [Shen & Win \(2010\)](#)

Fundamental Limits of Wideband Localization—Part I: A General Framework. IEEE TIT 56(10), 4956–4980.

S4 [Stanford STATS 200 · Lecture 14](#)

Consistency and asymptotic normality of the MLE.

EXTENSIONS ON BAILIPING.COM

Connect the ideas to inference and sensing.

[Radar SLAM](#) →

Sensing, motion, registration, and pose-graph correction.

[Frame registration](#) →

Geometry, alignment objectives, and observability.

[Variational inference](#) →

Approximation, optimization, and uncertainty.

Lab 1 · Sampling

Lab 2 · Bias

Lab 3 · Geometry

All demos run locally. No tracking, network requests, external fonts, or external JavaScript are required.